AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 18 · STUDENT LESSON

Error Analysis

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Keep the cases. Count every slice. Test the explanation.

Use error evidence to propose controlled follow-up without hiding regressions or claiming causes from associations.

Keep one habit: Complete aggregate comparison → pictured percent bars → thirty-case replay → paired-scene proposal → independent route change.

MAIN LESSON · SECTION 1

Which errors changed, and what would explain them?

> An error cluster suggests a test; it does not establish a cause.

Learning goals

I can categorize development errors using **evidence**.

I can design a controlled change with a stated **hypothesis**.

I can preserve failures and test boundaries.

Recall matrices and missing case context

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.
- Without opening the old lesson, explain one key idea from Precision and Recall. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Confusion Matrix. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Nearest Neighbors. Give an example and a mistake that the idea helps you avoid.
- What does a confusion matrix omit?

End goal

By the end, I can explain **error analysis** using a traced example and a stated limitation.

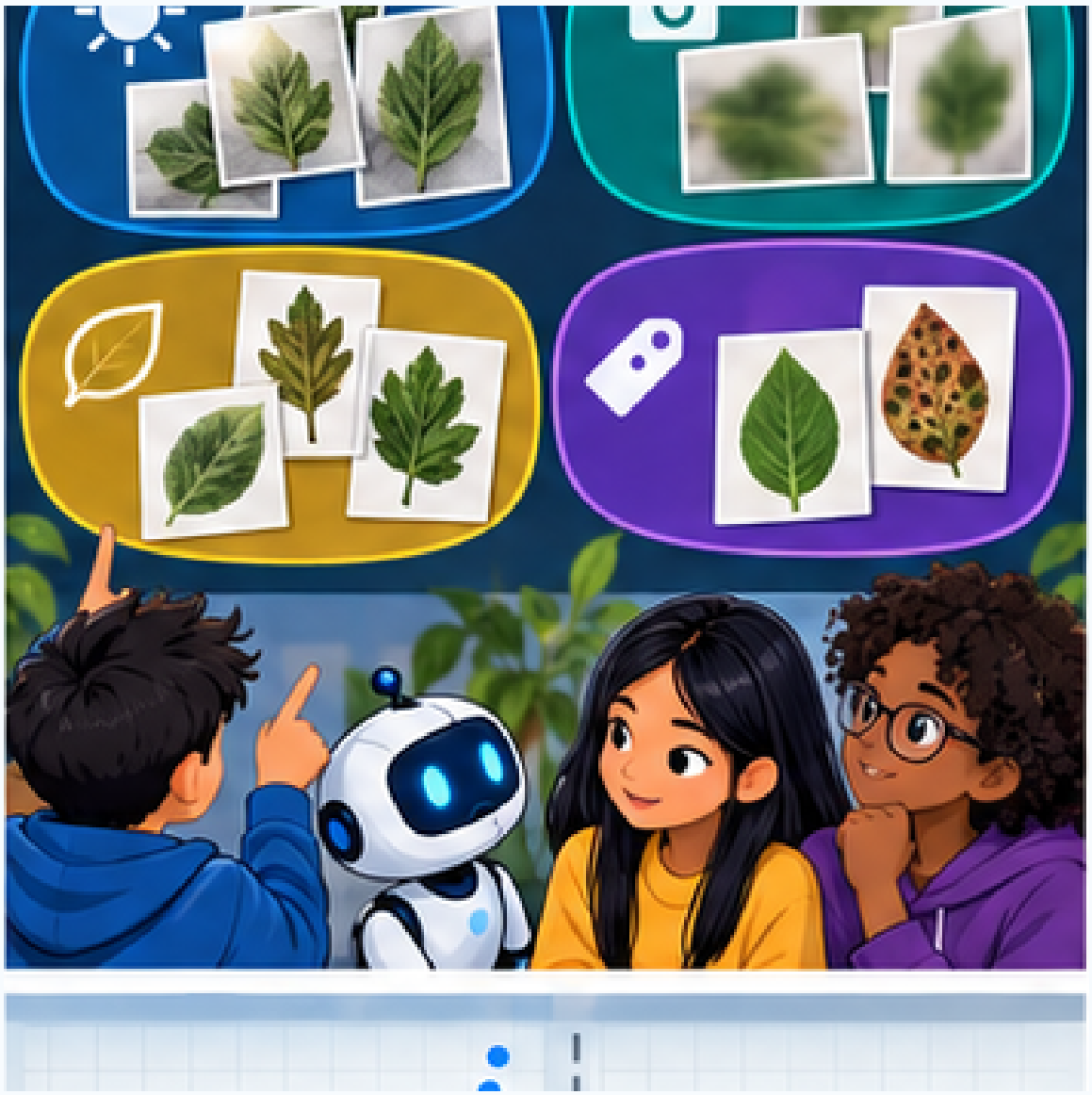
▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Grouping errors by observable conditions connects a metric problem to possible data, label, feature, or model causes.



LEARNING TARGET A model fails 6/10 backlit and 2/20 front-lit photos. Compute error rates and design a follow-up that tests lighting rather than leaf identity.

Builder name and experiment date:

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

Begin with IDs, references and collection conditions

> Keep the observations separate from proposed explanations.

Error-analysis purpose and fictional scope

Error analysis begins with **prediction** IDs and reference labels. Inspect features, collection conditions and the labelling policy. Do not assume every surprising **prediction** proves the reference is wrong.

Use development errors to propose a change and specify what outcome would support the explanation.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

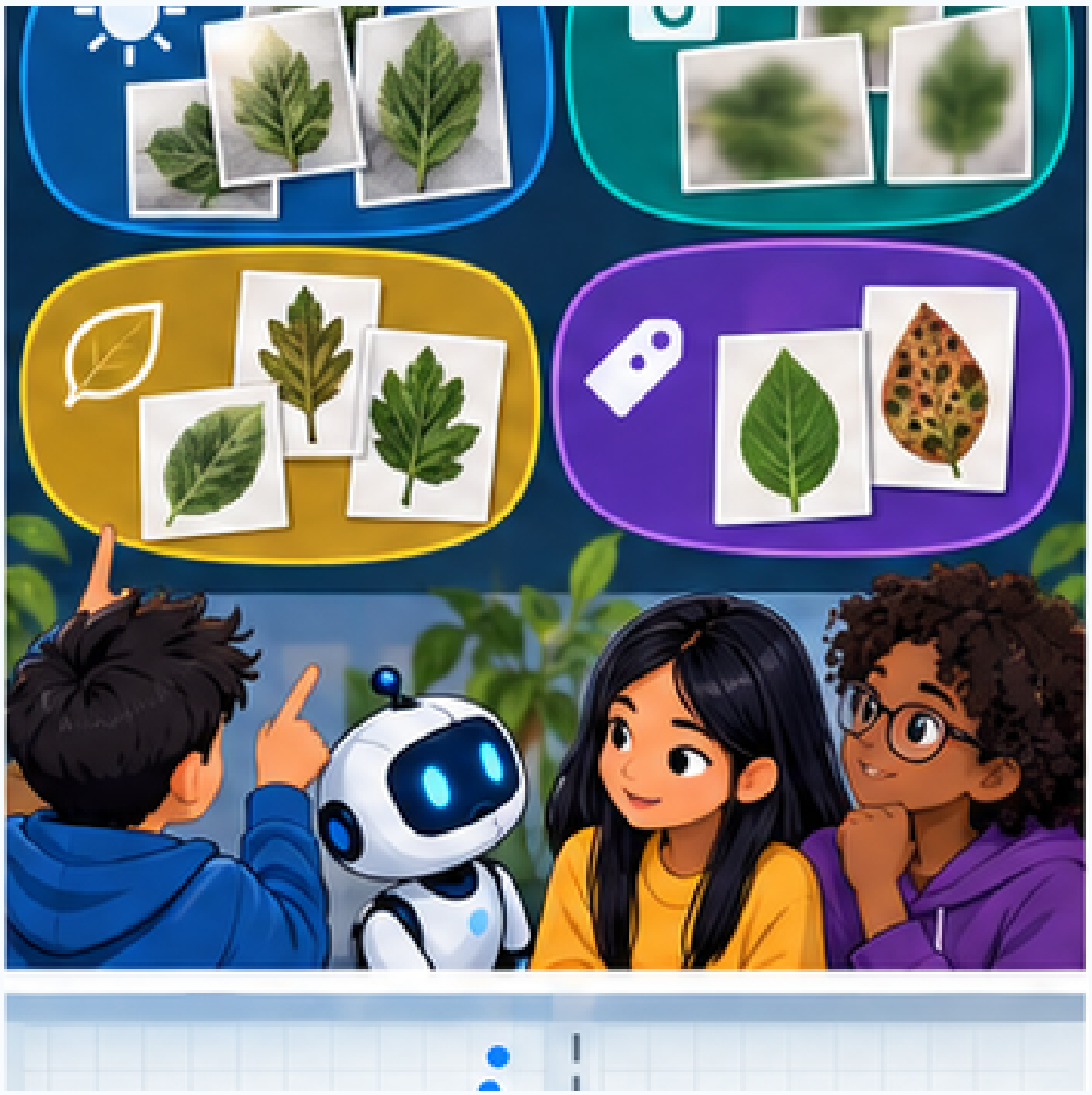
Earlier weeks supplied the model-building **process**; this week examines **error analysis** and controlled follow-up.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.
</details>



MYSTERY The team must use **error analysis** to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

MAIN LESSON · SECTION 3

Compare slice rates before and after a planned change

> A total improvement can conceal a regression in one group.

Complete aggregate development example

A validation audit has 12 dim-light cases with 4 errors and 18 bright-light cases with 3 errors. Rates are 4/12=1/3 and 3/18=1/6. Dim lighting is a useful lead, not proof of causation: camera, route or labels may also differ.

A planned change uses a **training**-fitted brightness normalization while keeping split and classifier settings fixed.

Optional complete supplied-count comparison script

```
old_errors = [4, 3]
new_errors = [2, 4]
counts = [12, 18]
for old, new, n in zip(old_errors, new_errors, counts):
    print(old / n, new / n)
print(sum(old_errors), sum(new_errors))
# Total errors fall from 7 to 6; one group worsens.
```

Trace before running

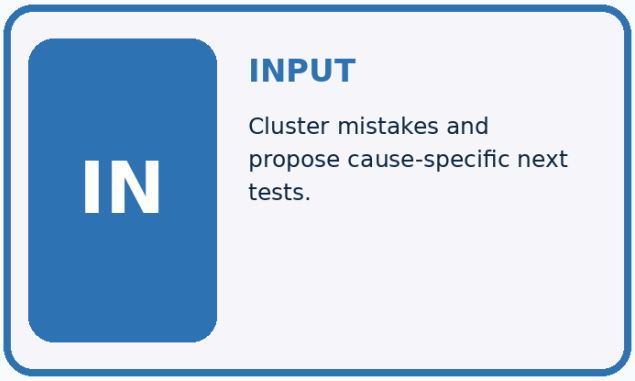
Trace the calculation on paper before running this optional Python check. Use the stated fictional **input**s and compare intermediate values. The code checks this example; your independent workbook case uses different **evidence**.

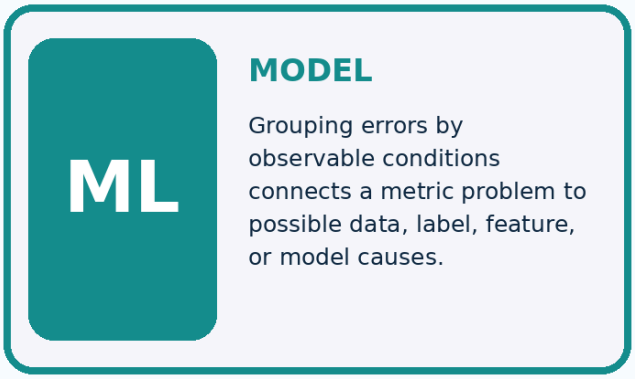
▼ **Earlier wording, alternatives and picture transcripts**

ML **MECHANISM**

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







QUESTION Cluster mistakes and propose cause-specific next tests.

MODEL IDEA Grouping errors by observable conditions connects a metric problem to possible data, label, feature, or model causes.

PROOF NEEDED Use an unseen case, visible calculation, **error analysis**, and a stated limitation.

UNDER THE HOOD Twelve errors are grouped: 6 low light, 3 blurred, 2 label disagreements, 1 unusual species. Largest cluster suggests a lighting representation problem, not proof of the cause.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Connect the picture and the explanation

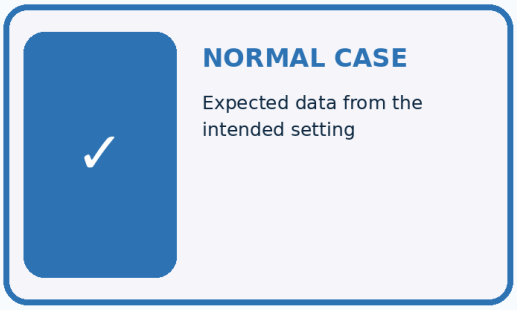
Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

MAIN LESSON · SECTION 4

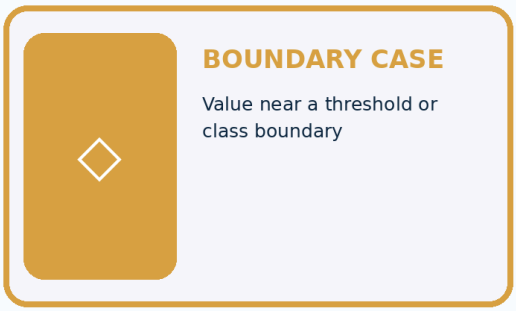
Compare an error lead with a controlled follow-up

> Count the cases in each condition before judging its error burden.

Four evaluation conditions



NORMAL Expected data and expected behavior.

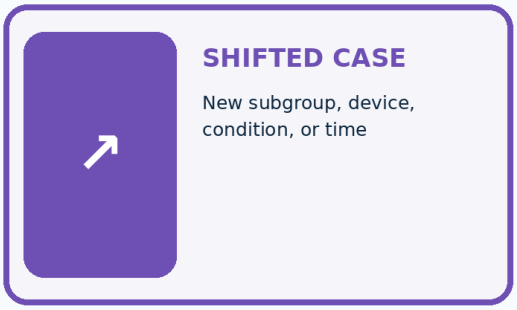


Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.



MISSING Necessary field, label, source, or measurement is unavailable.



SHIFTED New lighting, species, device, subgroup, or time condition.

Keep the claim bounded

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Choose a revealing case

- Which case most threatens today's claim, and why?

Predict the supplied changed case

- Change: Dim errors fall to1/6 in a later same-case replay.

▼ **Check after predicting**

Prediction: The replay's dim error rate is 16.7%; it is development **evidence** if those cases guided the revision.

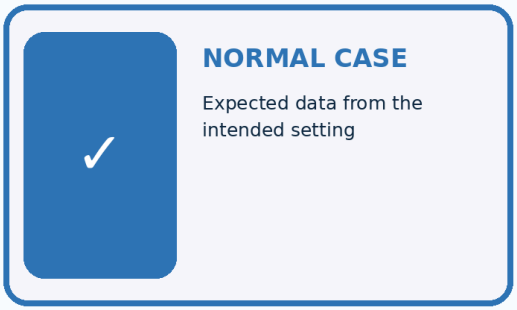
Reason: Observed improvement on reused cases is not an untouched final generalisation test.

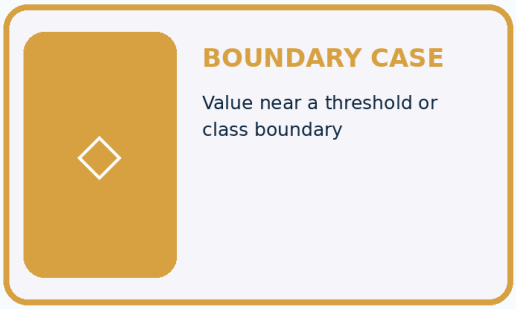
▼ Earlier wording, alternatives and picture transcripts

EVALUATION DESIGN

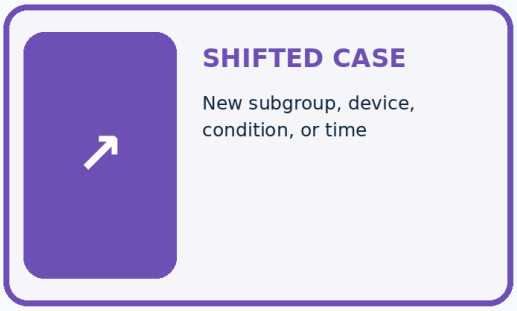
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Read slice denominators beside the error-rate bars

> Keep the main and pictured cohorts clearly separated.

Complete pictured facts and comparison scope

Find an error cluster before choosing a controlled follow-up.

Candidate errors: bright-light photos1/8; dim-light photos3/6. Images and reference labels are fixed for this audit.

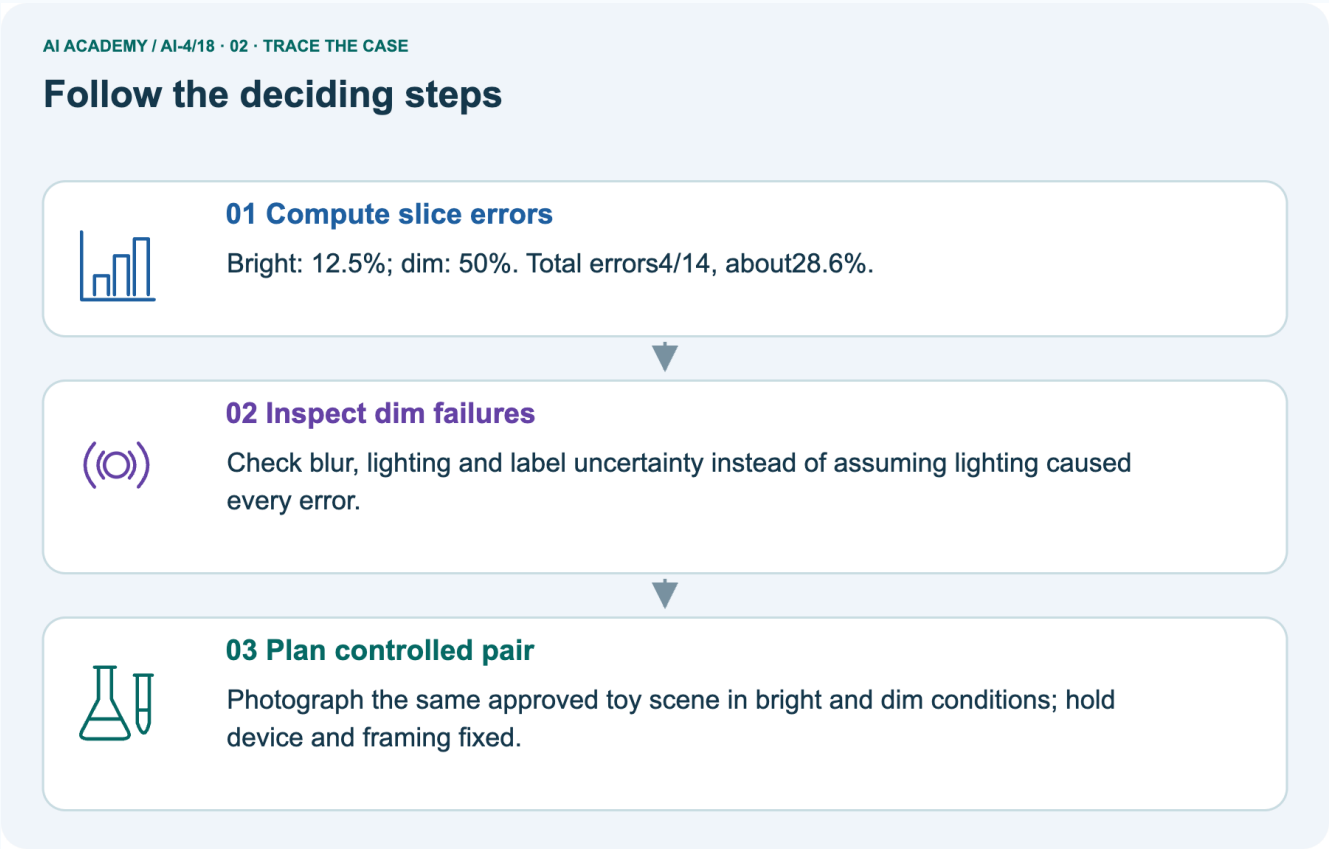
All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The main fixture has dim errors 4/12 and bright errors 3/18, later supplied as 2/12 and 4/18. Dim improves while bright worsens; total falls 7/30 to 6/30. The separate picture has bright 1/8 and dim 3/6, total 4/14. Its bar labels 12.5 and 50 are percentages, not counts from a

hundred measured photos. The pictured same-case replay changes dim to 1/6; it is reused development evidence, not fresh generalization.

None of these numbers alone proves lighting caused the errors.

Trace slice rates and a controlled-pair plan



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

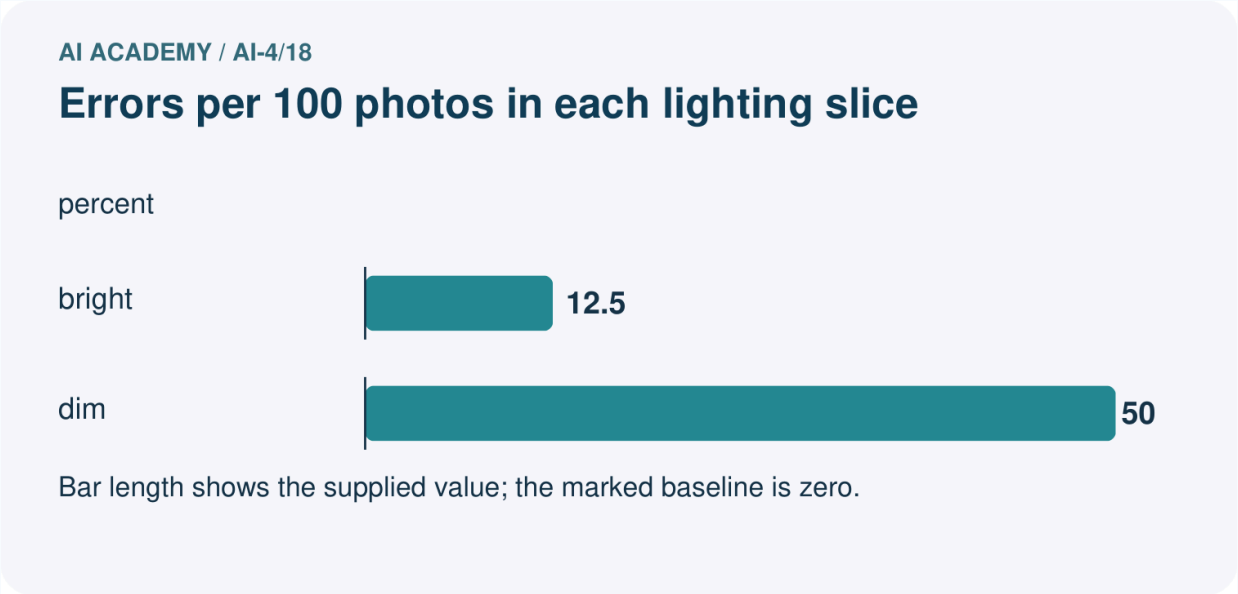
STEP	WHAT HAPPENS AND WHY
1. Compute slice errors	Bright: 12.5%; dim: 50%. Total errors4/14, about28.6%.
2. Inspect dim failures	Check blur, lighting and label uncertainty instead of assuming lighting caused every error.
3. Plan controlled pair	Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Inspect the original percent-scale bars

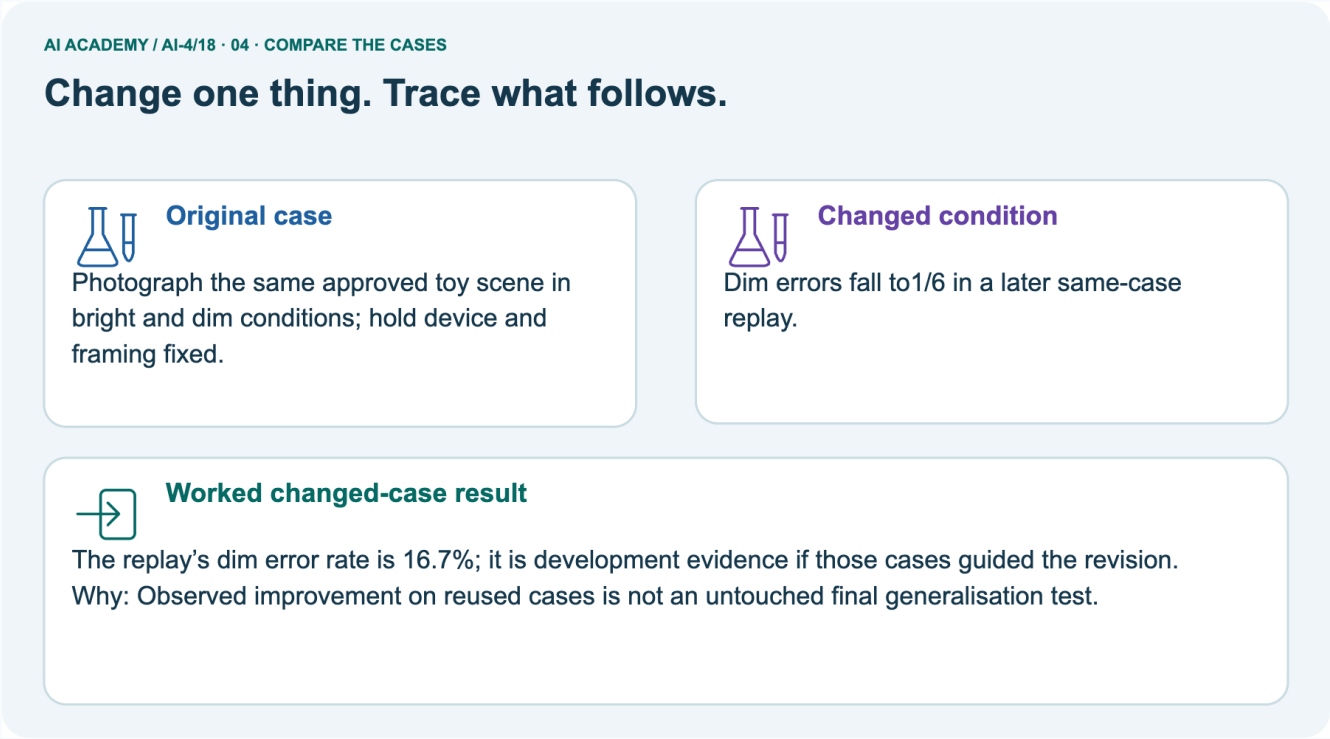


Follow the labelled picture with the matching explanation. Open the vector diagram

Predict a same-case dim replay

- Change one thing: Dim errors fall to1/6 in a later same-case replay. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain causal and fresh-case limits

- A slice difference suggests investigation; it does not establish a causal lighting explanation by itself.

- Read the labelled picture: What **evidence** would support the claim that a change helps fresh dim-light images? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Return to the main inspection question

Error analysis and controlled follow-up at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Inspect	IDs and conditions	Propose a hypothesis
Change	One planned factor	Keep comparison fixed
Evaluate	All validation records	Report gains and regressions

After the change, the same validation groups have 2/12 and 4/18 errors. Total errors fall from 7 to 6, but bright-light errors increase from 3 to 4. Report both results. A net improvement does not mean every part improved, and repeated experiments can overfit validation.

Does the difference prove dim light causes model errors? Explain the steps or evidence behind your response.

▼ **Earlier wording, alternatives and picture transcripts**

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 18 • WORKED EVIDENCE

1. Twelve errors are grouped: 6 low light, 3 blurred, 2 label disagreements, 1 unusual species.
2. Largest cluster suggests a lighting representation problem, not proof of the cause.
3. Collect or transform low-light examples, changing one factor.
4. Rerun the same evaluation and inspect other slices for regressions.

STEP 1 Twelve errors are grouped: 6 low light, 3 blurred, 2 label disagreements, 1 unusual species.

STEP 2 Largest cluster suggests a lighting representation problem, not proof of the cause.

STEP 3 Collect or transform low-light examples, changing one factor.

STEP 4 Rerun the same evaluation and inspect other slices for **regression**s.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

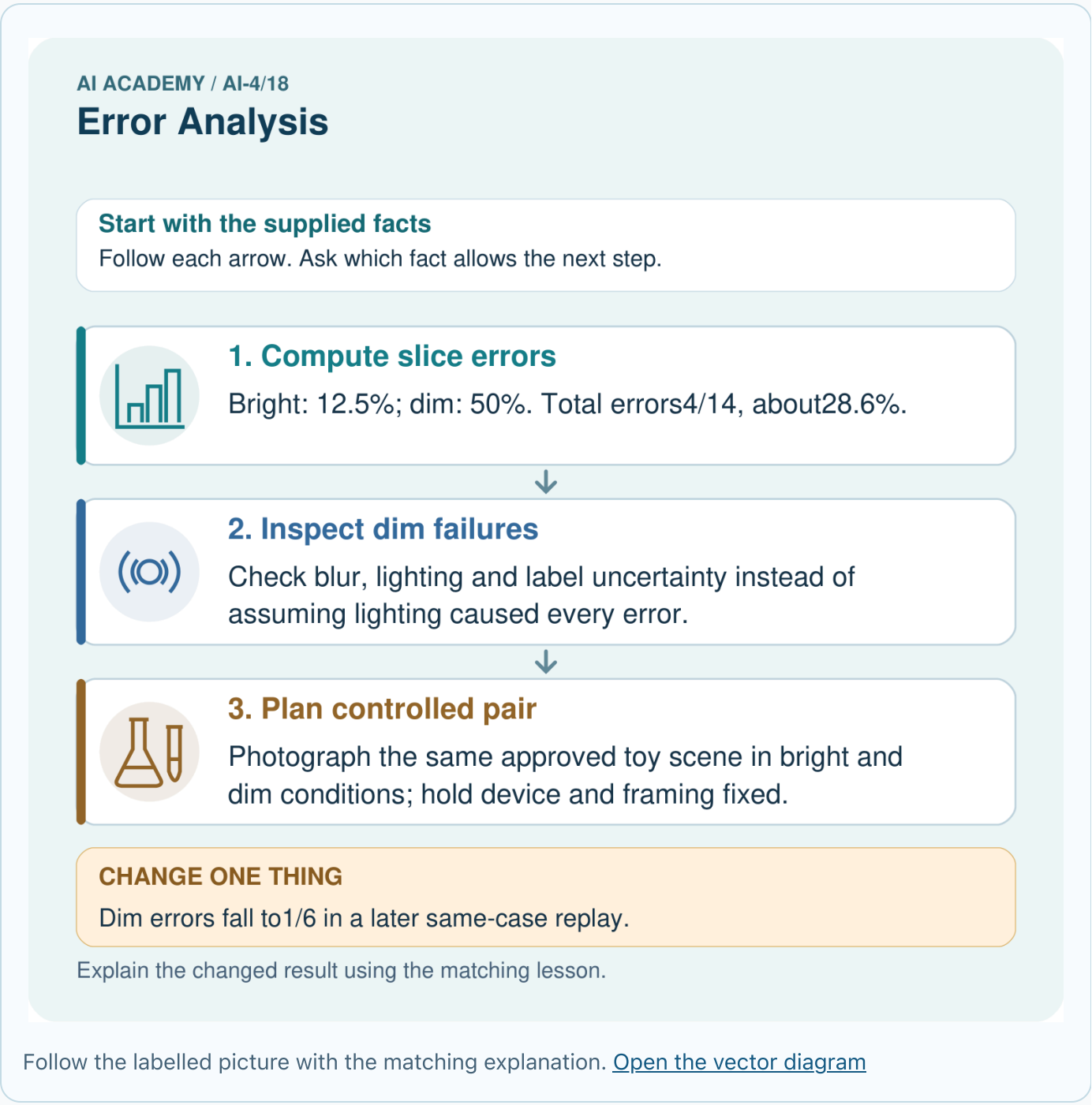
A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Error Analysis

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find a hypothesis reported as an established cause

> A surprising output does not automatically prove a bad reference label.

Diagnose the first claim

- Diagnose one cause and name the cleanest next experiment:

CLAIM TO CHALLENGE Looking closely at one surprising error is enough to diagnose the model.

State evidence and next-test limits

- Do not delete difficult cases to raise the score or reuse final-test errors for an unreported revision.

  Once test feedback guides a change, treat it as development **evidence** and obtain a new independent evaluation for a fresh final claim.

> **A team removes all disputed records only from the new model report. Why is the comparison invalid?**

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: Looking closely at one surprising error is enough to diagnose the model. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

A closer explanation

MAIN LESSON · SECTION 8

Practise a controlled change record with less help

> Report gains and regressions on the complete comparison set.

Try the supported limit question

Use workbook W2 for the connected example “**Error analysis** and controlled follow-up”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Save versions, predicted effects and all outcomes

Write a change record: **hypothesis**, exact code/data version, controlled factors, predicted effect, full before/after results and decision. Review disagreements using a consistent annotation policy.

If labels are corrected, version the data and rerun all candidates on the same corrected evaluation set.

Explain why evidence and procedure matter

- Use the worked case for **Error analysis** and controlled follow-up. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain what fresh dim-light evidence would require

- What **evidence** would support the claim that a change helps fresh dim-light images? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

Reserve the independent route comparison

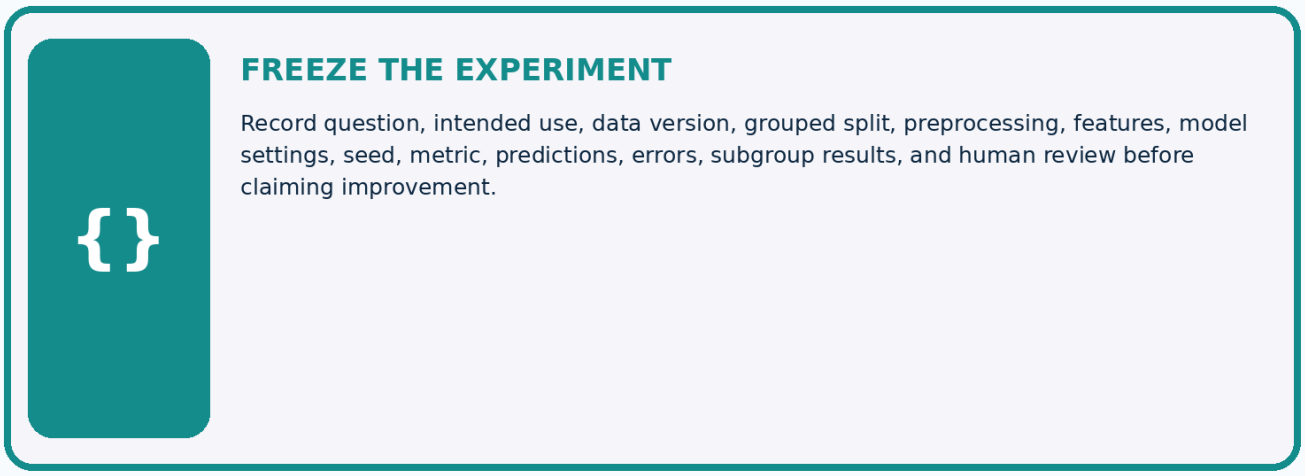
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Practise with less help: Explain the limit

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run Error Clinic with complete case-level evidence

> A useful next test must address the competing explanations, not just the biggest error count.

Complete thirty-case development replay

Case ID · Light / device / route · Reviewed reference · Old prediction · New supplied prediction

> **Case ID:** D01
> **Light / device / route:** dim / camera A / North
> **Reviewed reference:** clear
> **Old prediction:** blocked
> **New supplied prediction:** blocked

> **Case ID:** D02
> **Light / device / route:** dim / camera A / North
> **Reviewed reference:** blocked
> **Old prediction:** clear
> **New supplied prediction:** clear

> **Case ID:** D03

Light / device / route: dim / camera A / North

Reviewed reference: clear

Old prediction: blocked

New supplied prediction: clear

Case ID: D04

Light / device / route: dim / camera A / North

Reviewed reference: blocked

Old prediction: clear

New supplied prediction: blocked

Case ID: D05

Light / device / route: dim / camera A / North

Reviewed reference: clear

Old prediction: clear

New supplied prediction: clear

Case ID: D06

Light / device / route: dim / camera A / North

Reviewed reference: blocked

Old prediction: blocked

New supplied prediction: blocked

Case ID: D07

Light / device / route: dim / camera A / North

Reviewed reference: clear

Old prediction: clear

New supplied prediction: clear

Case ID: D08

Light / device / route: dim / camera A / North

Reviewed reference: blocked

Old prediction: blocked

New supplied prediction: blocked

Case ID: D09

Light / device / route: dim / camera A / North

Reviewed reference: clear

Old prediction: clear

New supplied prediction: clear

Case ID: D10

Light / device / route: dim / camera A / North

Reviewed reference: blocked

Old prediction: blocked

New supplied prediction: blocked

Case ID: D11

Light / device / route: dim / camera A / North

Reviewed reference: clear

Old prediction: clear

New supplied prediction: clear

Case ID: D12

Light / device / route: dim / camera A / North

Reviewed reference: blocked

Old prediction: blocked

New supplied prediction: blocked

Case ID: B01

Light / device / route: bright / camera B / South

Reviewed reference: clear

Old prediction: blocked

New supplied prediction: blocked

Case ID: B02

Light / device / route: bright / camera B / South

Reviewed reference: blocked

Old prediction: clear

New supplied prediction: clear

Case ID: B03

Light / device / route: bright / camera B / South

Reviewed reference: clear

Old prediction: blocked

New supplied prediction: blocked

Case ID: B04

Light / device / route: bright / camera B / South

Reviewed reference: blocked

Old prediction: blocked

New supplied prediction: clear

Case ID: B05

Light / device / route: bright / camera B / South

Reviewed reference: clear

Old prediction: clear

New supplied prediction: clear

Case ID: B06

Light / device / route: bright / camera B / South

Reviewed reference: blocked

Old prediction: blocked

New supplied prediction: blocked

Case ID: B07

Light / device / route: bright / camera B / South

Reviewed reference: clear

Old prediction: clear

New supplied prediction: clear

Case ID: B08

Light / device / route: bright / camera B / South

Reviewed reference: blocked

Old prediction: blocked

New supplied prediction: blocked

Case ID: B09

Light / device / route: bright / camera B / South

Reviewed reference: clear

Old prediction: clear

New supplied prediction: clear

Case ID: B10

Light / device / route: bright / camera B / South

Reviewed reference: blocked

Old prediction: blocked

New supplied prediction: blocked

Case ID: B11

Light / device / route: bright / camera B / South

Reviewed reference: clear

Old prediction: clear

New supplied prediction: clear

Case ID: B12

Light / device / route: bright / camera B / South

Reviewed reference: blocked

Old prediction: blocked

New supplied prediction: blocked

Case ID: B13

Light / device / route: bright / camera B / South

Reviewed reference: clear

Old prediction: clear

New supplied prediction: clear

Case ID: B14

Light / device / route: bright / camera B / South

Reviewed reference: blocked

Old prediction: blocked

New supplied prediction: blocked

Case ID: B15

Light / device / route: bright / camera B / South

Reviewed reference: clear

Old prediction: clear

New supplied prediction: clear

Case ID: B16

Light / device / route: bright / camera B / South

Reviewed reference: blocked

Old prediction: blocked

New supplied prediction: blocked

Case ID: B17

Light / device / route: bright / camera B / South

Reviewed reference: clear

Old prediction: clear

New supplied prediction: clear

Case ID: B18

Light / device / route: bright / camera B / South

Reviewed reference: blocked

Old prediction: blocked

New supplied prediction: blocked

Source, supplied-outcome status and confounding

Error Clinic source is a teacher-authored fictional development replay: twelve D cases are dim/camera A/North and eighteen B cases bright/camera B/South. These conditions co-vary, so their slice difference cannot isolate lighting. D01–D04 are old errors; D01–D02 remain new errors. B01–B03 are old errors; B01–B04 are new errors.

Old and new predictions are supplied fixtures, not the result of a model execution in this lesson. A planned training-fitted brightness change is a hypothesis; its training data and implementation are not supplied, so no fitting or causal success is claimed here.

Fully specified follow-up proposal

Follow-up field · Complete proposed classroom plan

Follow-up field: Question

Complete proposed classroom plan: Does changing only lighting alter errors for the fixed classifier on the same approved toy scenes?

Follow-up field: Pairs

Complete proposed classroom plan: Create new scene IDs S1–S4, each with a bright and dim view; keep subject/object positions, camera, route/background and framing fixed within a pair.

Follow-up field: Controls

Complete proposed classroom plan: Use the same frozen classifier and reviewed target rule; vary only the planned lighting condition and record collection order. Retain scene grouping for later source-separated evaluation.

Follow-up field: Evidence to record

Complete proposed classroom plan: Case/scene ID, condition, actual reference, frozen prediction, error, blur/visibility notes, and protocol deviations. No results are supplied for this proposal.

Follow-up field: Limits

Complete proposed classroom plan: Four scene pairs give limited evidence. If used to revise the model, they are development cases; evaluate the revised frozen method on fresh intended-use groups before a final claim.

Normal, regression, missing-context and changed-cohort probes

Complete error-analysis probe	Expected bounded check
Normal thirty-record replay	Old 7/30 errors; new 6/30; dim 4/12→2/12, bright 3/18→4/18.
Boundary one extra bright error B04	Record the regression even though the overall error count improves.

Complete error-analysis probe	Expected bounded check
Missing source/condition for proposed X01	Mark unknown metadata and investigate; do not assign dim or claim an explanatory slice rate from missing context.
Changed evaluation: replace bright cases with easier new scenes	Not a same-case before/after comparison; record changed cohort and do not attribute a difference only to the model update.

Inspect, hypothesize and plan

1. Annotate every wrong prediction using the complete D/B record table. Count old and new errors by slice and total, including the new B04 failure. Group observable conditions, propose at least one competing explanation and name which missing evidence would distinguish them. Design the supplied same-scene lighting comparison without inventing its outcomes.

   If a reference is reviewed and corrected, version it and compare both methods on the same corrected references while preserving the original evaluation record. Keep the independent route A/B task for an unaided first attempt.

Keep evidence and actual execution status

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____

> **Meaning / next test:** _____

> **Case:** Missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Shifted
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.
</details>

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Error Clinic: annotate every wrong prediction, cluster by condition, write hypotheses, and select one measurable next test.

1. Write the **hypothesis** and expected **evidence** before running.

2. Error Clinic: annotate every wrong **prediction**, cluster by condition, write hypotheses, and select one measurable next test.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: judge a change across two routes

> An improved target slice does not automatically mean overall improvement.

Complete independent route counts and scope

Old errors are 5/10 on route A and 2/20 on B. New errors are 3/10 and 5/20. State total and group changes and a justified next step.

This selected route task uses its own group sizes and old/new error counts, not the main lighting or pictured cohort. Keep A's denominator10 and B's20 for both methods. The two route groups partition this supplied30-case comparison; no cause, individual IDs or actual change implementation is provided. Report both aggregate and group directions before recommending a next test.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply independently

Week 18 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Of eight errors, six are on blurry images and two on clear images. Can you conclude blurry images have a higher error rate without group totals?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Create the Plant Classifier error log linking case ID, actual, predicted, confidence, condition, likely cause, and next action.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

A model fails 6/10 backlit and 2/20 front-lit photos. Compute error rates and design a follow-up that tests lighting rather than leaf identity.

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Keep the project error log linked to source evidence

> A next action should name the uncertainty it is intended to resolve.

Project artifact and evidence

THIS WEEK'S CHECKPOINT Create the Plant Classifier error log linking case ID, actual, predicted, confidence, condition, likely cause, and next action.

Keep case IDs, source groups, reviewed references, frozen predictions, score meaning if available, collection conditions, suspected explanations and planned tests. Preserve full old/new records, data/model versions and cohort membership. A likely cause is still a hypothesis.

Mark scores unavailable rather than invent confidence, report all regressions and reserve fresh groups for evaluating development-informed changes.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Limits, uncertainty and human decision

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

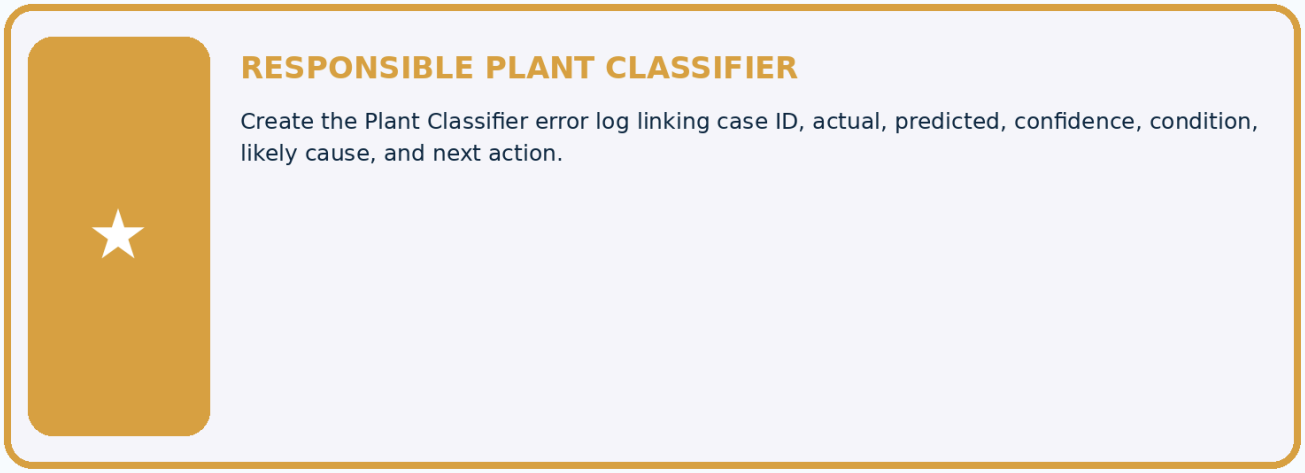
My **evidence**-linked project decision:

▼ **Earlier wording, alternatives and picture transcripts**

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



Evidence for your project

Finish with one observed regression and one testable explanation

> State what the record shows and what the proposed test has not yet established.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Categorize development errors using **evidence**

> **My deciding step, evidence and one remaining question**
>
> | |
> |---|
> | |
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Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the current experiment**

REFERENCE · SECTION 5

Reference · Words for the current experiment

> Tie each word to a particular input, fitted choice or evidence record.

Core vocabulary

Term	Precise meaning	My ML example
error slice	subset sharing a condition	
failure cluster	related mistakes with a likely cause	
root-cause hypothesis	testable explanation for errors	
controlled repair	one change aimed at one cause	

Closer-explanation terms

Word	Meaning
Error analysis	Inspection of mistakes to propose explanations
Hypothesis	A testable proposed explanation

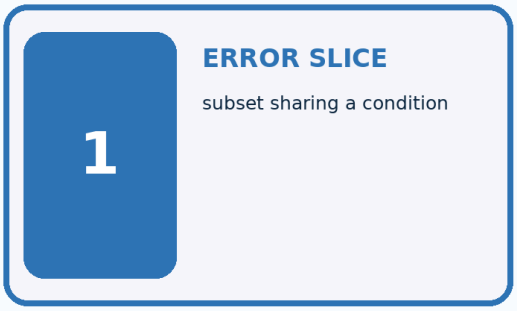
Word	Meaning
Controlled comparison	A comparison that changes one planned factor while holding relevant conditions fixed
Regression	Here: a new or worsened failure after an update. Next week the same word names numerical prediction, a different meaning

▼ Earlier wording, alternatives and picture transcripts

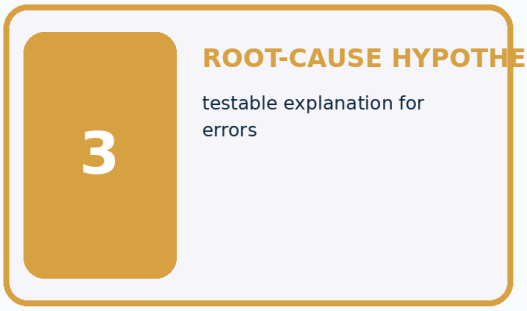
VOCABULARY

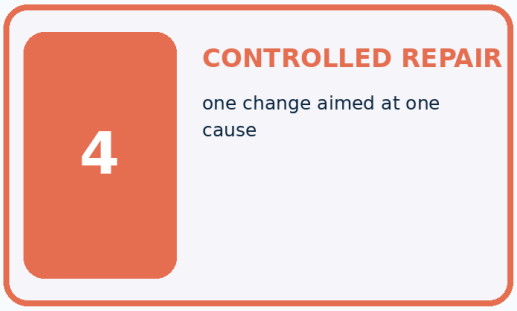
Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









MISCONCEPTION TO REPAIR Looking closely at one surprising error is enough to diagnose the model.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

▼ **Extra challenge · Build a stronger explanation**

REFERENCE · SECTION 10

Extra challenge · Build a stronger explanation

> Choose the next useful challenge after the main route.

Thinking ladder

- DEFINE Use all four terms precisely: **error slice**, **failure cluster**, **root-cause hypothesis**, **controlled repair**.
- CALCULATE Reproduce the worked **evidence** with units and denominator.
- TRACE Explain the data, code or decision path.
- DIAGNOSE Looking closely at one surprising error is enough to diagnose the model.
- TRANSFER Solve a fresh dataset or model case.
- IMPROVE Change one cause and rerun fixed **evidence**.
- DEFEND State intended use, metric, limitation, and human responsibility.

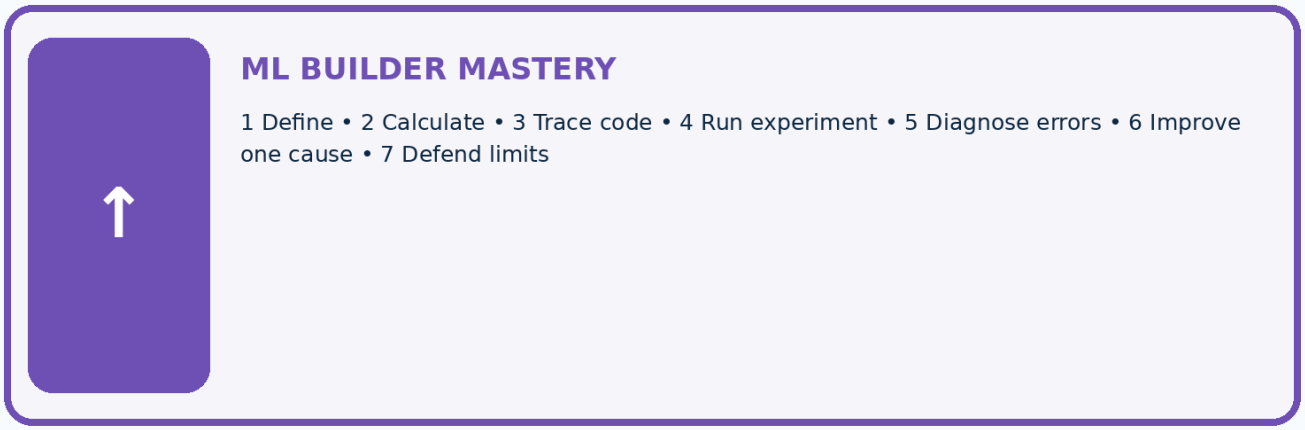
MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



▼ Reference · Inspect your evidence

REFERENCE · SECTION 12

Reference · Inspect your evidence

> Use after practice; keep the selected fresh route A/B before-after error comparison for the individual attempt.

Experiment evidence checklist

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

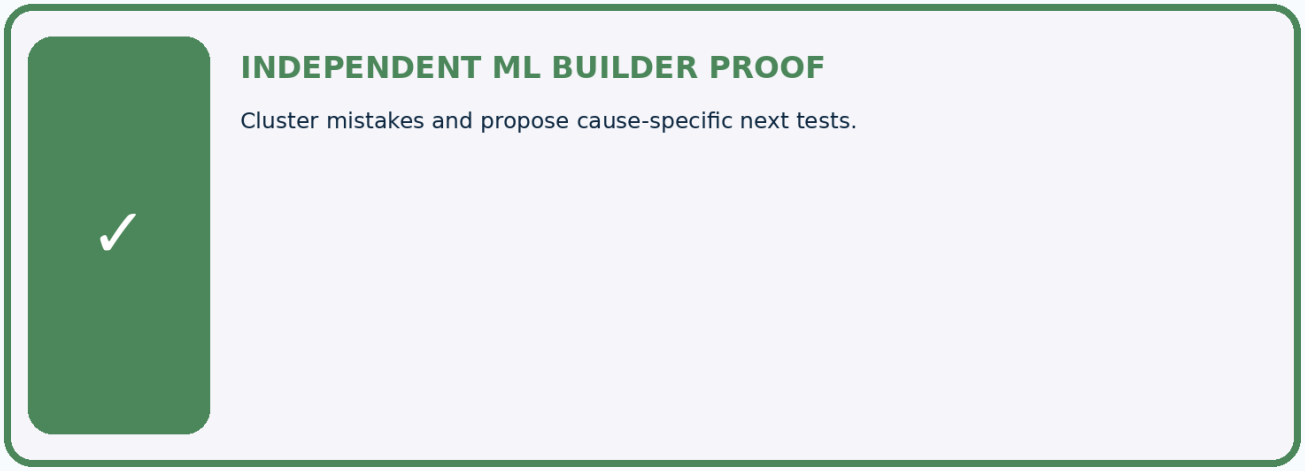
- Independent **evidence** and defense:
- My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



REQUIRED PROOF Cluster mistakes and propose cause-specific next tests.

NEXT BRIDGE Carry the experiment **evidence** into: **Regression** and Residuals.

▼ Extra practice · Earlier complete case

Extra practice · Earlier complete case

> Attempt the question before opening its worked explanation.

Situation and question

Four errors occur among five backlit photos; two among twenty ordinary photos.

Compare rates and investigate.

Your explanation

- Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

▼ Check after trying

Backlit error=**80%**; ordinary=10%. Inspect image quality and confounding differences.
Association motivates controlled testing; it does not alone prove cause.

Evidence or stage	Value or result
Backlit errors	4/5
Ordinary-light errors	2/20

▼ Earlier wording, alternatives and picture transcripts

Week 18 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Compare error rates, not only error counts

Four errors occur among five backlit photos; two among twenty ordinary photos.

Compare rates and investigate.

Backlit error=**80%**; ordinary=10%. Inspect image quality and confounding differences. Association motivates controlled testing; it does not alone prove cause.

▼ Reference · Earlier route checklist

REFERENCE · SECTION 15

Reference · Earlier route checklist

> Use one main route; revisit the checklist only when useful.

Checklist and first prediction

By the end, I can	How I will show it
Categorize development errors using evidence	Explain the deciding step, show a trace or test, and state one limit.
Design a controlled change with a stated hypothesis	Explain the deciding step, show a trace or test, and state one limit.
Preserve failures and test boundaries	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: Find an error cluster before choosing a controlled follow-up.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 17

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Find an error cluster before choosing a controlled follow-up.

- Which labelled step matches this detail? Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed.
- 1. Plan controlled pair
- 2. Compute slice errors
- 3. Inspect dim failures
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Plan controlled pair Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed. Read the picture in order.

Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: Dim errors fall to1/6 in a later same-case replay. Which conclusion is supported by the supplied facts?

- 1. The replay’s dim error rate is 16.7%; it is development **evidence** if those cases guided the revision.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: The replay’s dim error rate is 16.7%; it is development **evidence** if those cases guided the revision.

Observed improvement on reused cases is not an untouched final generalisation test.

Repair and keep the record

Review Precision and Recall (ai-4/17). Goal: Choose the denominator that answers the question. Short practice: Were the original precision and recall denominators the same?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Confusion Matrix (ai-4/15). Goal: Build a labelled confusion matrix from matched reference and **prediction** pairs. Short practice: Why is adding TP alone without reducing FN an incorrect revision?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Nearest Neighbors (ai-4/12). Goal: Compute absolute distances and rank **training** neighbours. Short practice: Why is the k=1 outcome at length5 not uniquely determined without a tie rule?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

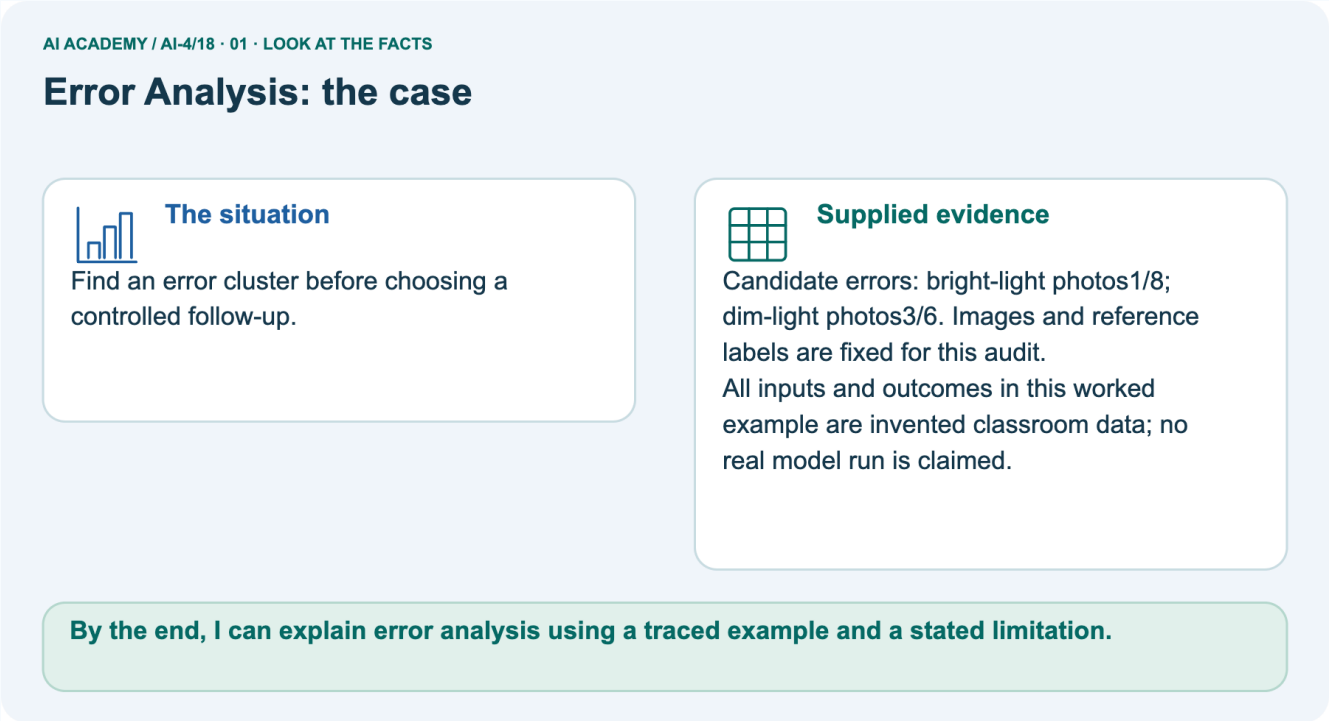
▼ **Reference · The six-picture journey**

REFERENCE · SECTION 18

Reference · The six-picture journey

> Choose the picture needed for the current question.

01. Error Analysis: the case



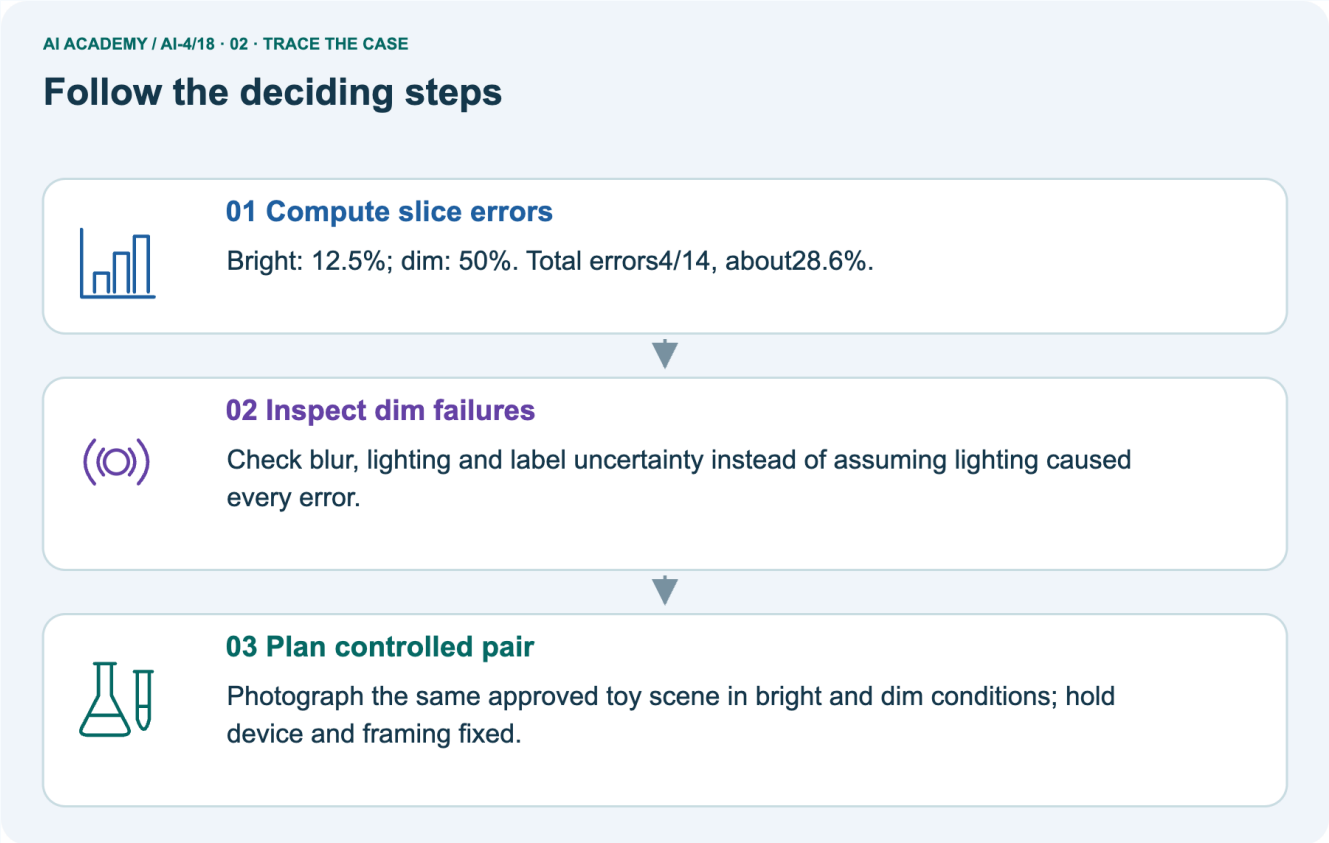
Follow the labelled picture with the matching explanation. Open the vector diagram

Find an error cluster before choosing a controlled follow-up. Supplied facts: Candidate errors: bright-light photos1/8; dim-light photos3/6. Images and reference labels are fixed for this audit.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



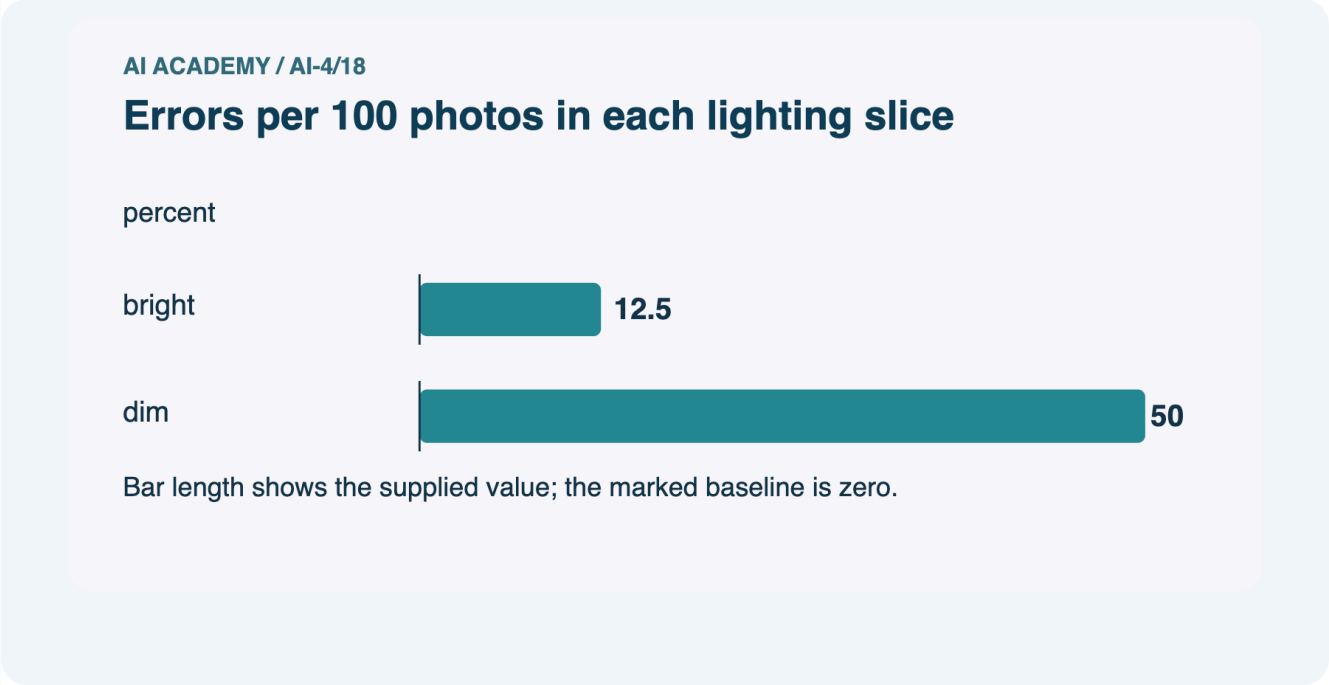
Follow the labelled picture with the matching explanation. Open the vector diagram

Compute slice errors: Bright: 12.5%; dim: 50%.

Total errors4/14, about28.6%. → Inspect dim failures: Check blur, lighting and label uncertainty instead of assuming lighting caused every error. → Plan controlled pair: Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed.

Point to each numbered stage and explain which recorded **input** it uses.

03. Errors per 100 photos in each lighting slice

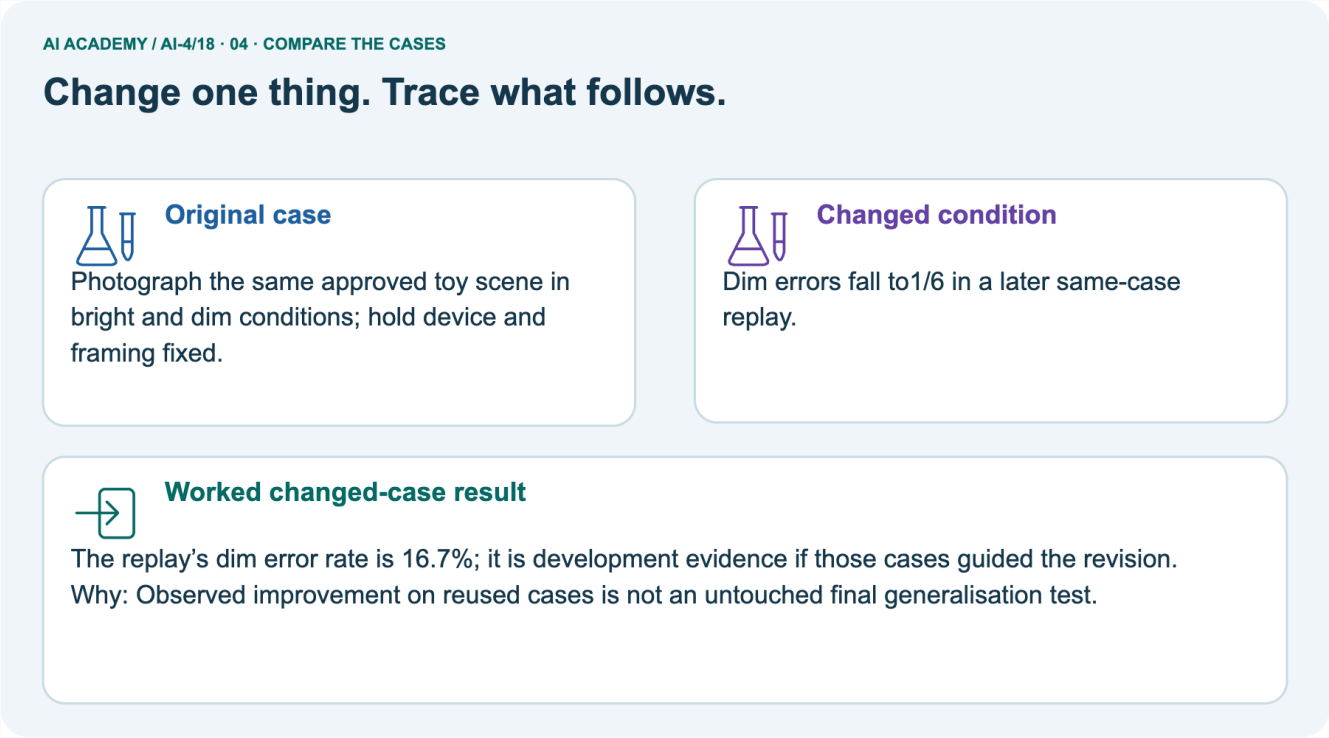


Follow the labelled picture with the matching explanation. Open the vector diagram

Errors per 100 photos in each lighting slice. Units: percent. bright: 12.5. dim: 50.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



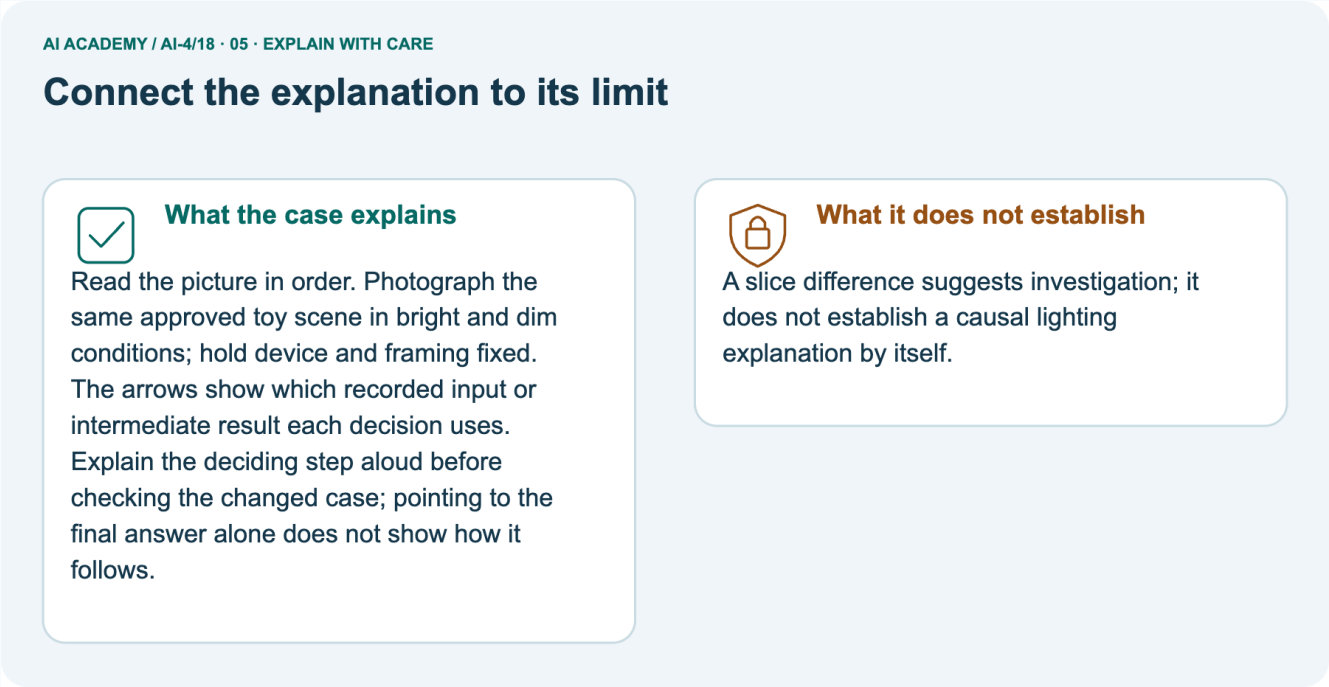
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed. Changed condition: Dim errors fall to1/6 in a later same-case replay.

Expected result: The replay's dim error rate is 16.7%; it is development **evidence** if those cases guided the revision. Reason: Observed improvement on reused cases is not an untouched final generalisation test.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

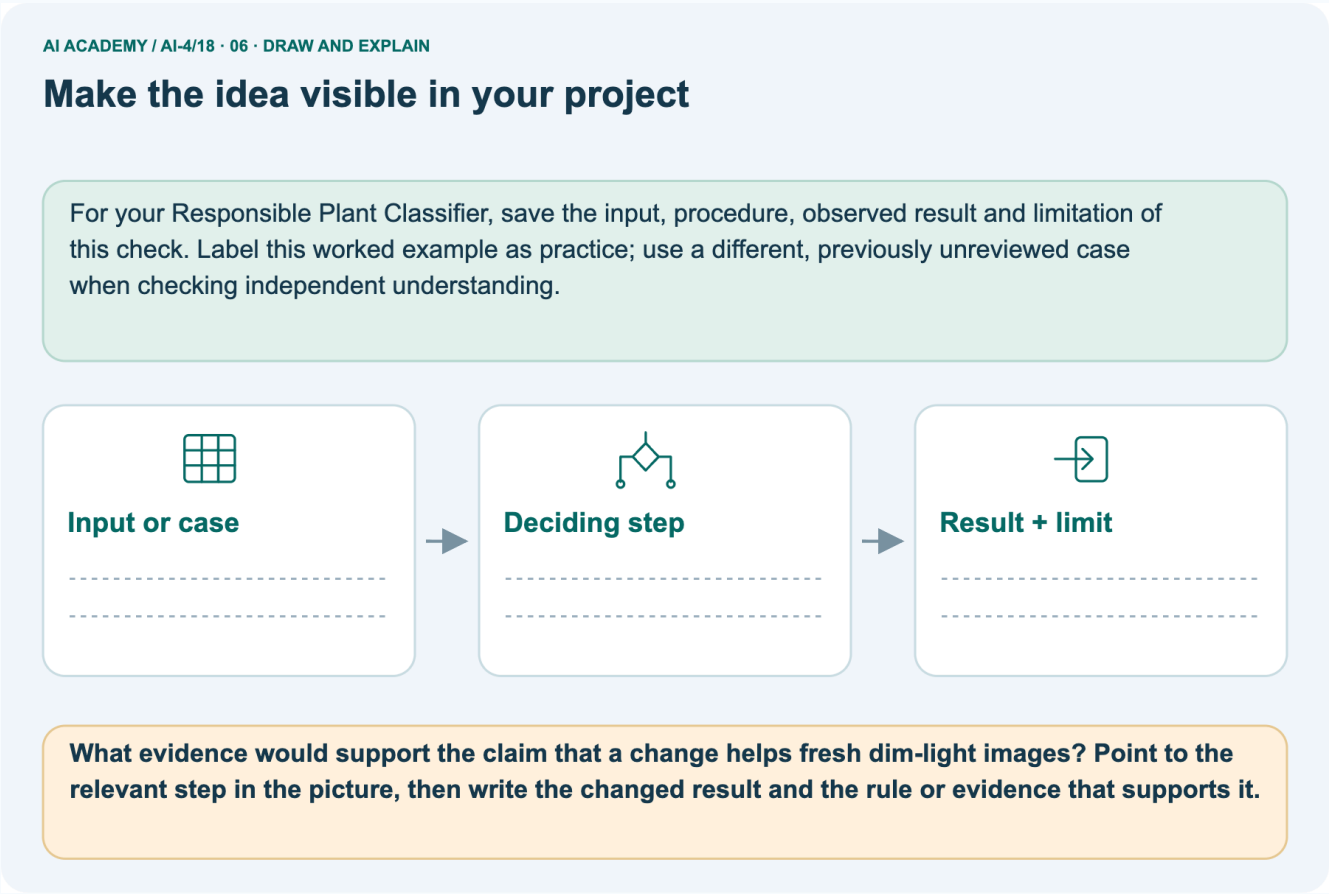
Read the picture in order. Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A slice difference suggests investigation; it does not establish a causal lighting explanation by itself.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What **evidence** would support the claim that a change helps fresh dim-light images? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Error Analysis: the case

02. Follow the deciding steps

03. Errors per 100 photos in each lighting slice

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.